

Roll No.

TCE-702

**B. Tech. (CE) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, 2019
TRANSPORTATION ENGINEERING—II
(RAILWAYS AND TUNNEL)**

Time : 1 : 30 Hours

Maximum Marks : 50

Note: (i) All questions are compulsory.

(ii) Answer any *two* sub questions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are ten.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

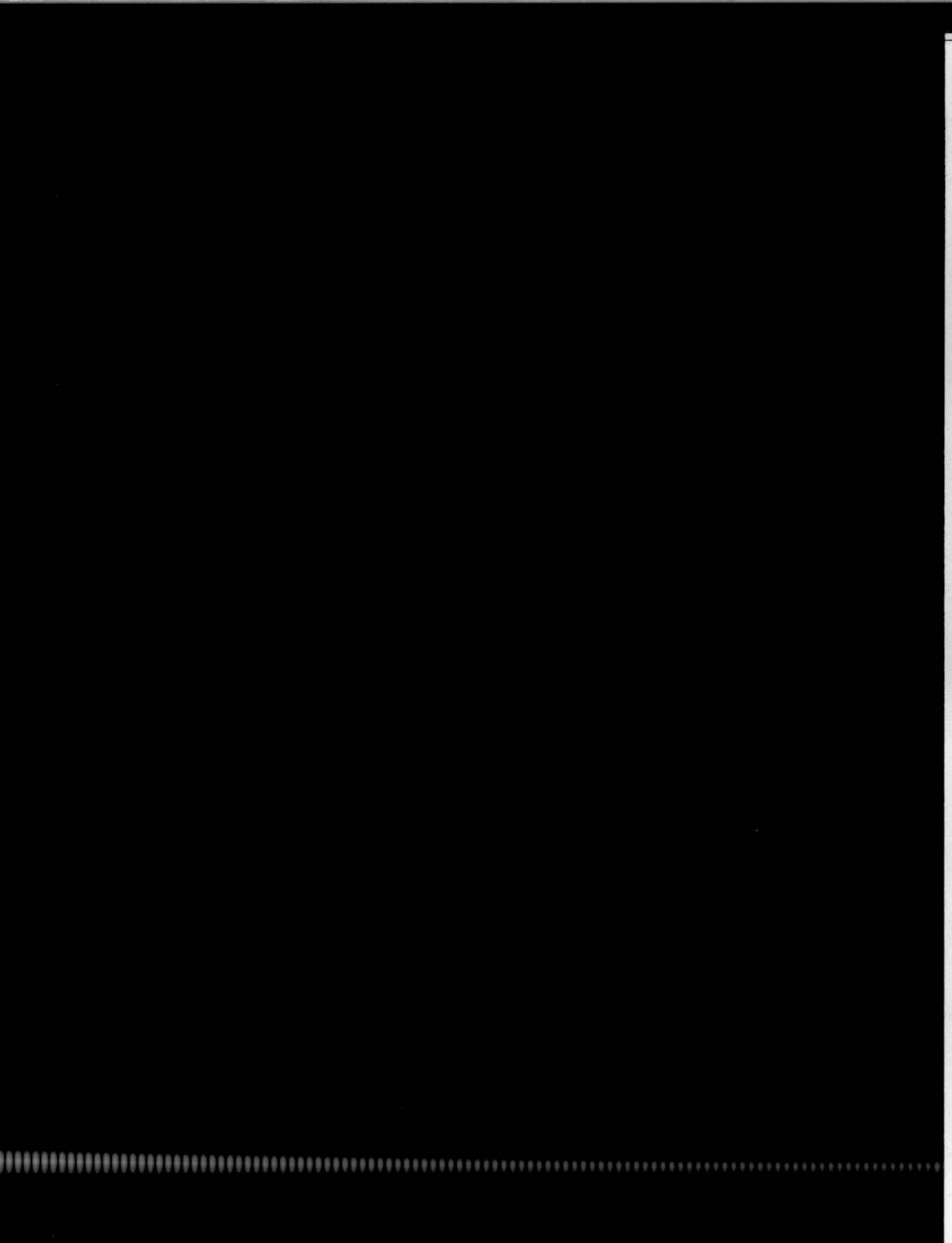
(a) Define gauge of railway track. Enumerate different gauges used in India. And discuss their suitability at different locations with their reasons.

(b) Why the uniformity of gauges is desirable in any country ? Mention the demerits of non-uniformity of gauges.

- (c) What do you understand by railway track ?
Draw the cross-section of different types of railway track.
2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Compare railway transport with highway transport.
- (b) Write a short note on permanent way and its components, with neat sketch.
- (c) What are the defects in Rails ? Explain.
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Discuss the necessity and effects of coning of wheels.
- (b) What is creep ? Discuss the theories propounded to explain probable causes of creep.
- (c) Discuss different types of rail joint with the help of neat sketches.
4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) A locomotive on MG track has four pairs driving wheels each carrying 20 tonnes. What maximum load can it pull on level track with curvature of 2.5° at 40 km/h ? (Assume suitable data if needed).

(3)

- (b) Explain the functions of sleepers in railway track. What are the desirable qualities or requirements of good sleepers ?
 - (c) Determine the number of sleepers required for the construction of 2000 m long BG railway track, ensuring the sleeper density of $(N + 5)$.
5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Explain, what is meant by track alignment. What are the basic requirements of good alignment ? What are the factors which control the alignment of a railway track ?
 - (b) What is the ballast in permanent way ? Mention the functions of ballast and state the requirements of a good ballast material.
 - (c) What are the essential requirements of fish plates ?



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TCE-703

**B. Tech. (CIVIL ENGINEERING)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, 2019**

CONSTRUCTION MANAGEMENT

Time : 1 : 30 Hours

Maximum Marks : 50

Note :(i) All questions are compulsory.

(ii) Answer any *two* sub questions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are ten.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

(a) Discuss the role of management in project execution.

(b) Describe two phases of project management.

(c) Explain risk cost management.

(2)

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2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Discuss the main causes of project failure.
 - (b) Explain Capital Investment proposal and Net present value.
 - (c) What is Gantt Bar Chart ?
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Explain all the functions of project management.
 - (b) Explain work breakdown structure.
 - (c) Explain in brief the importance of project scheduling.
4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Explain in detail the various project management techniques.
 - (b) Explain in brief the different elements of network.
 - (c) Write a short note on concept of precedence network analysis.

(3)

5. Attempt any *two* parts of choice from (a), (b) and (c).
(2×5=10 Marks)

- (a) Explain the principles for developing an organization for effective and efficient working.
- (b) Write a short note on Staffing and Benefit cost ratio.
- (c) Give all the objectives of construction project management.

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TCE-712

**B. Tech. (CIVIL ENGINEERING)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, 2019
GEOGRAPHICAL INFORMATION SYSTEM**

Time : 1 : 30 Hours

Maximum Marks : 50

Note :(i) All questions are compulsory.

(ii) Answer any *two* sub questions among (a), (b) and (c) in each main question.

(iii) Total marks for each main question are **ten**.

1. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
 - (a) Give the standard definition of GIS. Also give the objectives of GIS.
 - (b) The data for GIS collected from different sources. Explain in brief all the data sources for GIS.
 - (c) Explain the History of GIS in the world and its development in India.

(2)

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2. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) What are the Errors in GIS ? Explain in brief.
 - (b) Explain GIS as per NCGIA and ESRI.
 - (c) Explain in brief Terrain Data from Satellite Remote Sensing.
3. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Define Map projection and Geo Referencing.
 - (b) Define Edge Matching and Rubber Sheeting.
 - (c) Explain Quad-wee Tessellation and Block Coding.
4. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)
- (a) Explain the concepts model of spatial information with the neat diagram.
 - (b) Explain the application of GIS in different fields.
 - (c) Write a short note on Scanner and Digitizer.

(3)

5. Attempt any *two* parts of choice from (a), (b) and (c). (2×5=10 Marks)

- (a) What are the errors may arise during capturing of spatial and attribute data ?
- (b) Define Data Representation and explain all the variables of Data Representation.
- (c) Explain in detail all the essential components of GIS.